Week 10 — Quantum Sensing: Limits, Fisher Information, and Nanophotonics

EEQT30001 Quantum Technology and Engineering, Unit 2 (Quantum Optics and Quantum Sensing)

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Learning objectives (Week 10)

1. Derive, completely, the shot-noise phase limit of an interferometer: Poisson statistics → error propagation → $\Delta\phi = 1/\sqrt{\langle n\rangle}$; distinguish the standard quantum limit ($1/\sqrt{N}$, statistics) from the Heisenberg limit ($1/N$, entanglement), and explain how LIGO injects (frequency-dependent) squeezed vacuum at its dark port.
2. Use the quantum Fisher information: state the Cramér–Rao bound (result only), derive $F_Q = 4(\Delta G)^2$ for pure states, and work the two canonical examples — coherent state $F_Q = 4\langle n\rangle$ (SQL) versus N00N state $F_Q = N^2$ (Heisenberg), including why loss disarms the N00N advantage.
3. Articulate the design thesis of nanophotonic quantum sensing: the generator G is set by light–matter coupling, the coupling by the vacuum mode structure, and the mode structure by the nanostructure — so the quantum limit is a *design parameter*; map today's frontier (maximum-information states, Fisher-information flow) and its open quantum structure-end and emitter-end problems.

Reading: Degen, Reinhard & Cappellaro, Rev. Mod. Phys. **89**, 035002 (2017), Secs. I–II and the tables only. Shot-noise pieces: QO_photon_stat (Poisson pages); QO_note_6, Sec. 2.4.

Opening puzzle. LIGO must resolve a strain of order 10^{-21}: over its 4-km arms, a length change of $\sim 4\times10^{-18}$ m ($\approx 1/400$ of a proton diameter; the popular "1/10,000" figure refers to LIGO's best displacement sensitivity, $\sim 10^{-19}$ m). Today's lecture asks a single question about that feat: *who sets the endpoint of measurement precision — Heisenberg, or statistics?* The answer (statistics, far more often than you think) is the key that unlocks everything else: why squeezed light helps, what Fisher information measures, and why the precision limit itself can be engineered.

1 Shot Noise and the Standard Quantum Limit

1.1 Poisson statistics: the noise of independence

A laser beam delivers photons *independently*: the arrival of one tells you nothing about the next (Week 7: $g^{(2)}(0) = 1$). For independent events the photon count n in a fixed window follows the Poisson distribution

$$P(n) = e^{-\langle n\rangle}\frac{\langle n\rangle^n}{n!}, \qquad (\Delta n)^2 = \langle n\rangle \;\Rightarrow\; \Delta n = \sqrt{\langle n\rangle}. \tag{1.1}$$

This $\sqrt{\langle n\rangle}$ is *shot noise*. In a photocurrent the same statistics read $\langle \Delta i^2\rangle = 2e\,\Delta f\,\langle i\rangle$ — white noise proportional to the mean current. Nothing quantum has been assumed beyond graininess plus independence — a pairing worth remembering for the end of the lecture.

1.2 The complete derivation: $\Delta\phi = 1/\sqrt{\langle n\rangle}$

This is the only derivation in Unit 2 that we carry through every step — because every sensor in this lecture lives or dies by it. Send a coherent state $|\alpha\rangle$ into a Mach–Zehnder interferometer with a phase ϕ in one arm (Fig. 1); the second input port carries vacuum. Week 8's beam-splitter algebra (QO_note_6, Sec. 2.4) gives coherent states at both outputs, with intensities

$$I_1 = |\alpha|^2 \sin^2\frac{\phi}{2}, \qquad I_2 = |\alpha|^2 \cos^2\frac{\phi}{2}, \qquad S(\phi) \equiv \frac{I_2 - I_1}{|\alpha|^2} = \cos\phi. \tag{1.2}$$

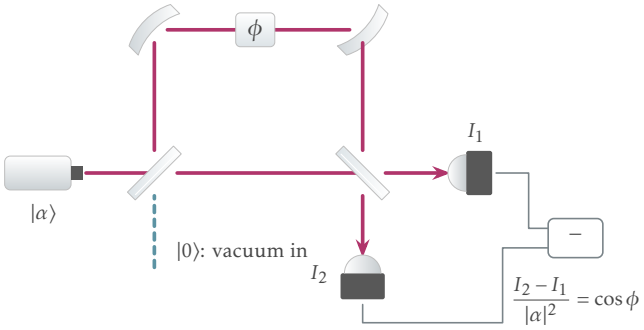


Figure 1: Phase readout with a Mach–Zehnder interferometer. The unused input port is not idle: its vacuum fluctuations are the noise floor of the measurement.

Now the error propagation, done honestly:

Step 1 — what fluctuates. The detected counts n_1, n_2 are independent Poisson variables with means I_1, I_2 (in photons per window), so the difference $D = n_2 - n_1$ has

$$\mathrm{Var}(D) = \mathrm{Var}(n_1) + \mathrm{Var}(n_2) = I_1 + I_2 = |\alpha|^2 = \langle n\rangle. \tag{1.3}$$

Note the sines and cosines add to one: the noise is *independent of* ϕ even though the signal is not.

Step 2 — noise on the normalized signal. $\Delta S = \Delta D/\langle n\rangle = \sqrt{\langle n\rangle}/\langle n\rangle = 1/\sqrt{\langle n\rangle}$.

Step 3 — from signal noise to phase noise. A fluctuation ΔS masquerades as a phase fluctuation $\Delta S/\left|dS/d\phi\right|$ with $dS/d\phi = -\sin\phi$:

$$\Delta\phi = \frac{\Delta S}{|\sin\phi|} = \frac{1}{\sqrt{\langle n\rangle}|\sin\phi|} \xrightarrow[\phi=\pi/2]{} \boxed{\Delta\phi_{\mathrm{SQL}} = \frac{1}{\sqrt{\langle n\rangle}}} \tag{1.4}$$

The interferometer is operated at the half-fringe $\phi = \pi/2$, where the slope is steepest. Equation (1.4) is the *standard quantum limit* (SQL). Nothing in the derivation invoked an uncertainty relation: the $1/\sqrt{N}$ came from Eq. (1.3), i.e., from adding the variances of *independent* events.

1.3 SQL versus Heisenberg

Read $1/\sqrt{N}$ for what it is: the central limit theorem. N independent photons are N independent polls of the same phase, and averaging independent polls improves as $\sqrt{N}$ — the same law that governs election surveys. The way past it is therefore not a better laser but a *correlated* probe: if the N photons are entangled they can act as a single object that accumulates phase N times faster, and the ultimate scaling allowed by quantum mechanics is the *Heisenberg limit* $\Delta\phi = 1/N$ [8] (Sec. 2 makes this precise). Between the two limits lies this week's slogan, seeded here: *when you beat the SQL, you are not beating Heisenberg — you are beating statistical independence.*

1.4 LIGO: replace the vacuum, lower the noise

Where does the shot noise of Eq. (1.3) physically enter a Michelson interferometer? Through the *dark port* (Fig. 2) — the output kept on a dark fringe, where "nothing" exits and, by unitarity, vacuum enters. LIGO sits just off the dark fringe and reads the signal against the carrier as a local oscillator (DC readout): in the shot-noise limit this gives the same $1/\sqrt{\langle n\rangle}$ scaling as the half-fringe readout of Sec. 1.2, and the photodiode never sees the full laser power. The dark line of this unit closes here: the vacuum is not empty (Week 7), and its fluctuations beating against the strong carrier are precisely the noise floor of Eqs. (1.3)–(1.4). Caves' 1981 insight [1]: nothing says the dark port must carry *ordinary* vacuum. Inject squeezed vacuum $S(r)|0\rangle$ (Week 7's resource, produced by Week 8's parametric amplifier) with the squeezed quadrature aligned to the phase readout, and the homodyne signal $I_1 - I_2 = \sqrt{2}\,|\alpha|\langle X_\theta\rangle$, with $X_\theta = (ae^{-i\theta} + a^\dagger e^{i\theta})/\sqrt{2}$ and α the local-oscillator amplitude (in LIGO, the carrier),[1] carries less noise than the SQL. GEO600 has run squeezed light continuously since 2010 [2]; Advanced LIGO since the O3 run (2019), with ≈3 dB of shot-noise reduction (2.7 dB typical, 3.2 dB maximum) [4]; the 2013 LIGO result [3] was a demonstration.

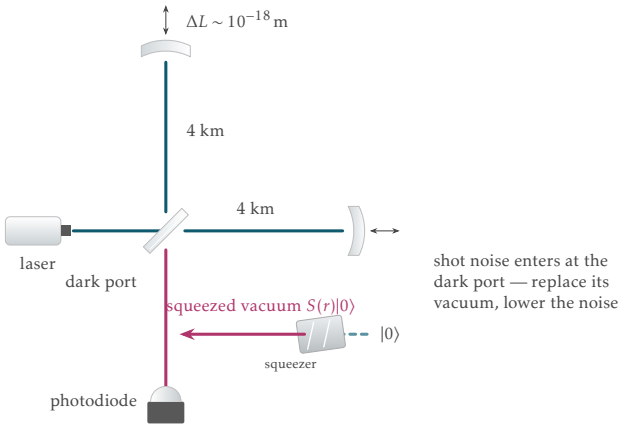


Figure 2: The dark port is where the vacuum — and therefore the shot noise — enters a Michelson interferometer. Replacing that vacuum with squeezed vacuum is how LIGO measures below the SQL.

Frequency-dependent squeezing. One subtlety earns its own paragraph (and an exam-bank slot). Squeezing the phase quadrature necessarily *anti*-squeezes the amplitude quadrature, which pushes on the mirrors: radiation-pressure noise, dominant at low frequency, gets *worse*. LIGO therefore rotates the squeezing angle with frequency — a 300-m filter cavity reflects the squeezed vacuum with a frequency-dependent phase — so that high frequencies see phase squeezing (less shot noise) while low frequencies see amplitude squeezing (less radiation-pressure noise) (Fig. 3). This is running in O4 [5]: quantum noise reduced across the whole band, not traded from one end to the other.[2]

In-class discussion 1. *You double the laser power. How much does the phase sensitivity improve?* By $\sqrt{2}$: Eq. (1.4) scales as $1/\sqrt{\langle n\rangle}$. That square-root stubbornness — 100 times the power for

[1] Gerry & Knight's 1/2 quadrature convention (Sec. 7.3) gives $2|\alpha|\langle X(\theta)\rangle$.

[2] Two "SQLs" circulate: ours, Eq. (1.4), is the coherent-state shot-noise limit; the gravitational-wave community's *free-mass* SQL is the optimum envelope of shot plus radiation-pressure noise. Fixed-angle squeezing cannot beat the latter; frequency-dependent squeezing can (first sub-SQL: Yu *et al.* [6]).

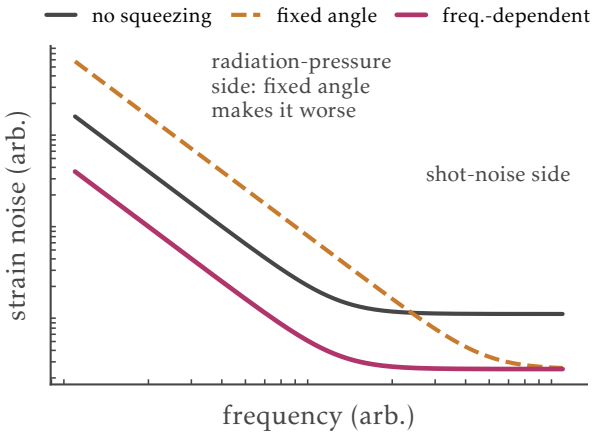


Figure 3: Why the squeezing angle must rotate with frequency (schematic noise budget, after Ref. [5]). Fixed-angle squeezing lowers shot noise but raises radiation-pressure noise; frequency-dependent squeezing lowers both.

10 times the precision — is why LIGO chases squeezing instead of ever-bigger lasers (and why radiation pressure punishes the brute-force route anyway).

Summary: Section 1

Independent arrivals ⇒ Poisson ⇒ $\Delta n = \sqrt{\langle n \rangle}$, Eq. (1.1). Propagating that noise through the MZI fringe gives the unit's one complete derivation: $\Delta\phi_{\mathrm{SQL}} = 1/\sqrt{\langle n \rangle}$ at the half-fringe, Eq. (1.4) — a statement about statistics, not about Heisenberg. The noise enters at the dark port, where only vacuum lives; LIGO replaces that vacuum with squeezed vacuum, rotated with frequency (filter cavity) so shot noise and radiation-pressure noise fall together.

2 Parameter Estimation and Quantum Fisher Information

2.1 The estimation framework, and the bound we do not prove

Every sensor is the same abstract machine: a parameter θ (phase, field, displacement) is imprinted on a probe state, $|\psi_0\rangle \to |\psi_\theta\rangle$; a measurement produces data; an estimator turns data into $\hat{\theta}$. Classical statistics assigns the data a Fisher information (FI) $F(\theta)$ and bounds any unbiased estimator by the *Cramér–Rao* inequality — after ν repetitions,

$$\Delta\theta \geq \frac{1}{\sqrt{\nu F(\theta)}}. \tag{2.1}$$

We take Eq. (2.1) as given (result only; it is theorem, not physics). The quantum step [7]: maximizing F over *all* measurements permitted by quantum mechanics defines the *quantum Fisher information* (QFI) $F_Q \geq F$ — the information content of the state itself, before any detector choices.

2.2 Pure states: $F_Q = 4(\Delta G)^2$

For this course's scope — pure states, unitary encoding $|\psi_\theta\rangle = e^{-i\theta G}|\psi_0\rangle$ with Hermitian generator G — the QFI collapses to one line. Expand the overlap of neighboring states:

$$\langle\psi_\theta|\psi_{\theta+d\theta}\rangle = \langle e^{-i\,d\theta\,G}\rangle = 1 - i\,d\theta\,\langle G\rangle - \tfrac{1}{2}d\theta^2\langle G^2\rangle + \mathcal{O}(d\theta^3), \tag{2.2}$$

so the overlap probability is $|\langle\psi_\theta|\psi_{\theta+d\theta}\rangle|^2 = 1-(\Delta G)^2 d\theta^2+\cdots$. Distinguishability of neighboring states *is* information about θ; the identification $|\langle\psi_\theta|\psi_{\theta+d\theta}\rangle|^2 \equiv 1 - \frac{1}{4}F_Q\,d\theta^2$ gives

$$\boxed{F_Q = 4(\Delta G)^2} \qquad \Longrightarrow \qquad \Delta\theta \;\geq\; \frac{1}{2\,\Delta G\,\sqrt{\nu}}. \tag{2.3}$$

The rule of the whole subject in five words: *sensitivity is generator variance.* A state estimates θ well iff it is spread widely over the eigenvalues of whatever operator generates θ.

2.3 Example 1 (blackboard): coherent state — the SQL again

Phase encoding on one mode: $G = \hat{n}$, probe $|\alpha\rangle$. Week 7 (and QO_note_3, Sec. 2.2): a coherent state has Poissonian number statistics, $(\Delta n)^2 = \langle n\rangle$. Therefore

$$F_Q = 4\langle n\rangle, \qquad \Delta\phi \;\geq\; \frac{1}{2\sqrt{\langle n\rangle}}. \tag{2.4}$$

The $1/\sqrt{\langle n\rangle}$ scaling of Sec. 1 reappears — now as a statement about the *state*, independent of any interferometer bookkeeping (constant factors depend on the setup; the scaling does not). Classical light cannot scale better, because Poisson variance is what independence looks like in the number basis.

2.4 Example 2 (blackboard): N00N state — the Heisenberg limit

Take the two-mode "all-or-nothing" state and phase encoding in arm a:

$$|\mathrm{N00N}\rangle = \frac{|N,0\rangle + |0,N\rangle}{\sqrt{2}}, \qquad G = \hat{n}_a. \tag{2.5}$$

Then $\langle G\rangle = N/2$ and $\langle G^2\rangle = N^2/2$, so $(\Delta G)^2 = N^2/4$ and

$$F_Q = N^2, \qquad \Delta\phi \;\geq\; \frac{1}{N}. \tag{2.6}$$

Heisenberg scaling, achieved: the N photons behave as one object of N times the frequency (the interference fringe oscillates as $\cos N\phi$ — "super-resolution"). The bonus notebook, problem 3, verifies Eq. (2.6) numerically in QuTiP for $N = 2, 4, 6$. Figure 4 compares the two scalings.

In-class discussion 2. *If N00N states reach $1/N$, why does LIGO inject squeezed vacuum instead?* Loss. Lose a single photon from a N00N state and the environment learns which arm held all N — the superposition collapses and *all* phase information dies. At LIGO-scale losses the N00N advantage is gone entirely, while squeezed vacuum degrades gracefully (a lossy squeezed state is a less-squeezed state). Fragility is the price of maximal $(\Delta G)^2$; practical quantum sensing is the art of buying variance without buying fragility.

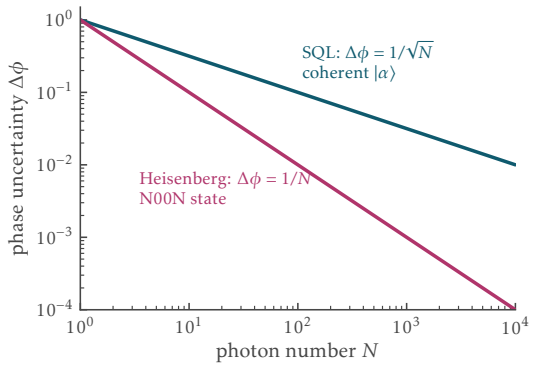


Figure 4: The two scalings of phase estimation, Eqs. (2.4) and (2.6). The gap between them is what "quantum-enhanced sensing" means.

2.5 Platforms at a glance

The framework of Eq. (2.3) is platform-blind — pick a generator, engineer its variance [8]:

platform	**quantity (generator)**	**quantum resource**
NV centers in diamond	magnetic field (spin $\hat{S}_z$)	long-lived spins, room T
trapped ions / lattice clocks	time–frequency ($\hat{H}$)	entangled ions, 10^{-18} clocks
atom interferometers	gravity, acceleration ($\hat{p}$)	matter-wave coherence
LIGO / optomechanics	displacement, strain ($\hat{n}$)	squeezed vacuum
photonic probes	phase, refractive index ($\hat{n}$)	N00N, squeezed, single photons

Summary: Section 2

Estimation is bounded by Cramér–Rao, Eq. (2.1) (result only); maximizing over measurements defines F_Q. For pure states with unitary encoding, $F_Q = 4(\Delta G)^2$, Eq. (2.3): sensitivity *is* generator variance. Coherent states give $F_Q = 4\langle n\rangle$ (the SQL, restated as a property of the state); N00N states give $F_Q = N^2$ (the Heisenberg limit) but die with one lost photon — which is why LIGO runs on squeezed vacuum instead. The generator, and hence the limit, is chosen by the physics of the platform.

3 Nanophotonics: The Limit Is a Design Parameter

3.1 From my research to this question

One sentence of history, then only the future. My own work — topological photonic lasers, non-Hermitian systems, quantum thermal radiation — has been one long variation on a single question: *how does structure rewrite the rules that light and matter must obey?* This section argues that measurement precision is the next rule to be rewritten. Two research lines have already proven, classically, that the precision limit is not a constant of nature but a *contract between the wave and the structure it lives in.* They optimized the light. What remains — and what this lab intends to do — is to optimize the *structure* for the quantum Fisher information, and the *quantum emitters that live inside it.*

The bridge from Sec. 2 is Week 7's oldest friend. The QFI is generator variance, Eq. (2.3); for an emitter-based sensor the generator is built from the light–matter coupling; and the coupling is set by the vacuum mode structure at the emitter — which is exactly what a cavity redesigns. The Purcell factor $F_P = \frac{3}{4\pi^2}(\lambda_0/n)^3\, Q/V$ [9] was this course's first proof that the vacuum is a design material (Fig. 5; recall Week 7's density-of-states figure). Chain the dependencies:

$$F_Q \longleftarrow \text{coupling } g \longleftarrow \text{vacuum mode structure} \longleftarrow \text{nanostructure design}, \tag{3.1}$$

and the conclusion reads itself: *the quantum limit is a design parameter.* The dark line of the unit ends here — the vacuum is not empty, it is raw material, it computes, it can be squeezed — and designing it is nanophotonics' home turf.

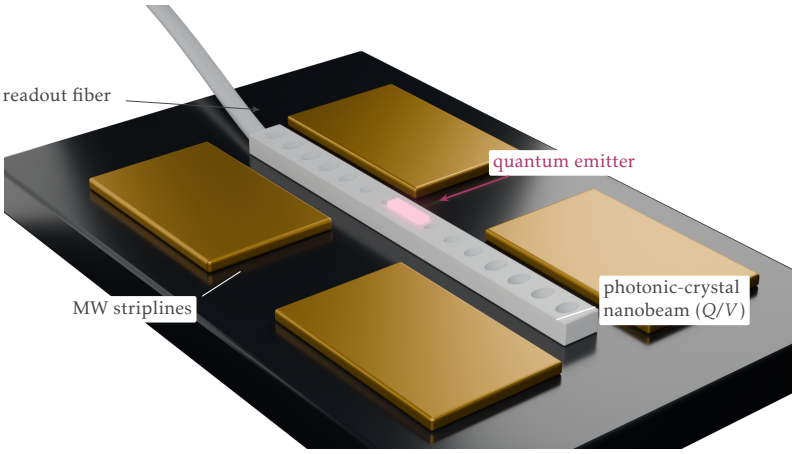


Figure 5: The kind of object this section is about: a quantum emitter inside a photonic-crystal nanobeam cavity, whose designed vacuum sets the coupling — and, through Eq. (3.1), the achievable F_Q. Microwave striplines drive the emitter's spin; a fiber collects the photons.

3.2 Act I — two classical lines, each half of the story

The Zheludev line: topological light as a picometric ruler. Deeply subwavelength features of structured light — superoscillations, phase singularities, optical skyrmions — vary faster than any Fourier component of the field, and that steepness is metrological gold: a natural ruler with tick marks far finer than the wavelength [10]. The "optical ruler" measured nanometric displacements this way [11], and topologically structured scattering plus deep learning has since pushed localization to the picometre scale, below thermal fluctuations — *picophotonics* [12]. Half the story: the ruler is exquisite, but the analysis is classical.

The Rotter line: Fisher information as an operator. For any linear scattering system, ask: which input field extracts the most information about a parameter θ buried in the medium? The answer has closed form (Sec. 3.4): the Fisher information is the expectation value of a Hermitian operator built from the scattering matrix, and the optimal probe is its top eigenvector — the *maximum-information state* [13], demonstrated to boost precision by an order of magnitude behind a disordered medium. Two more results give the program its shape: the FI, averaged over scattering environments, is *invariant* [14] — so any enhancement must come from *design*, never from complexity itself — and the FI density obeys a Poynting-like continuity equation: information *flows*, and can be steered by the medium [15]. Half the story: the formalism is general, but it optimizes the light — the structure is given, usually disordered.

3.3 Act II — convergence, and the first quantum step

The two lines have begun to merge: a Southampton series on the *flow* of Fisher information in picometric optical metrology, one paper of which is joint with the Rotter group (Weimar *et al.*, arXiv:2508.13640, preprint). The quantum step has also been taken — as a tutorial: from S and $\partial_\theta S$ one can build the quantum operators whose optimization jointly shapes the spatial profile *and* the quantum state of the probe [16]. In parallel, quantum light has learned to survive complex media: adaptive optics with entangled photons now works [17, 18]. Look at the map, though (Fig. 6): everything above sits in the *state-end* column — given a (usually disordered) structure, optimize what you send in. The ancestry is distinguished — multimode squeezed beams beat the SQL for beam displacement in 2003 [19], and a change of measurement basis dissolved Rayleigh's curse in 2016 [20] — but it is all the same move: *optimize the light.*

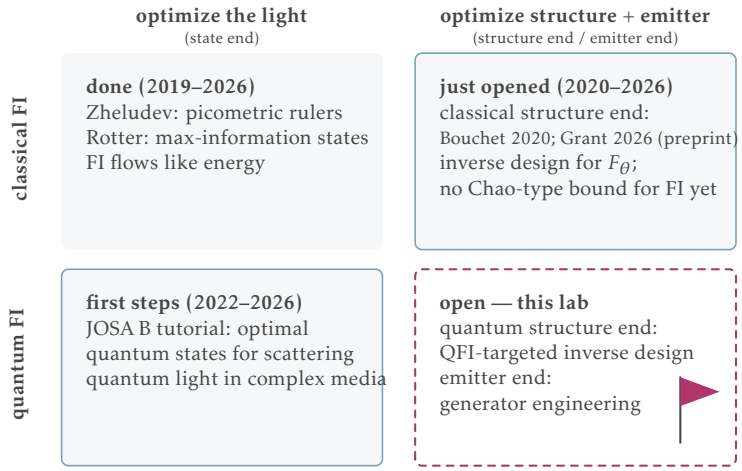


Figure 6: The state of the field, as a map. Both classical lines and the quantum first steps optimize the *light*; the classical structure end has just opened; the quantum structure end and the emitter end (flag) are this lab's claim.

3.4 Act II$\frac{1}{2}$ — the toy model: $F_\theta = \partial_\theta S^\dagger \partial_\theta S$

The core of the Rotter program fits on half a page — classical linear algebra, no quantum mechanics needed. A linear structure maps incoming mode amplitudes $\mathbf{c}$ (normalized, $\mathbf{c}^\dagger\mathbf{c} = 1$) to outgoing ones, $\mathbf{c}_{\text{out}} = S(\theta)\mathbf{c}$, where θ is the parameter to estimate (a position, a refractive index, a displacement). How strongly does the output depend on θ?

$$\|\partial_\theta \mathbf{c}_{\text{out}}\|^2 = \mathbf{c}^\dagger \underbrace{\partial_\theta S^\dagger \partial_\theta S}_{\equiv F_\theta} \mathbf{c} = \langle F_\theta \rangle_{\mathbf{c}}\,. \tag{3.2}$$

For shot-noise-limited coherent detection the Fisher information is, up to a detection prefactor, exactly this expectation value [13]. F_θ is Hermitian and positive semi-definite, so the best probe is pure eigen-analysis: *the maximum-information state is the top eigenvector of* F_θ, and its eigenvalue is the information budget (Fig. 7). All of wavefront-shaping metrology is Eq. (3.2) plus hardware. And now the question this section exists to ask: **what is the quantum version of this eigenvalue problem — when the probe is not a wavefront but an emitter inside the structure, and the object being designed is not c but S itself?** Ref. [16] is the first step — and it still optimizes the state.

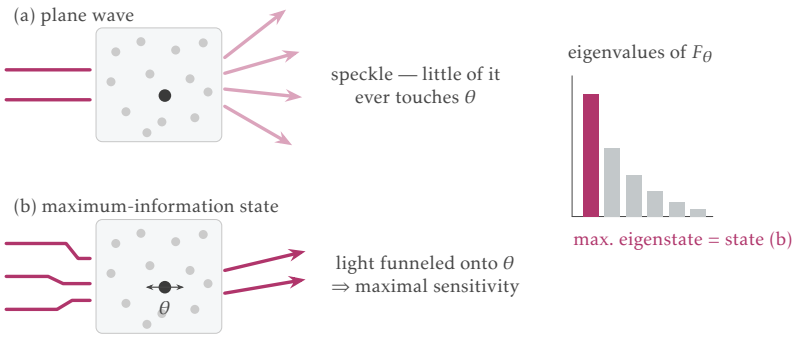


Figure 7: The toy model, Eq. (3.2) (after Ref. [13]). A plane wave wastes most of its light on modes that never touch the parameter θ; the top eigenvector of F_θ funnels everything onto it.

3.5 Act III — the manifesto, and three thesis seeds

Here is the claim. The local density of states (LDOS) and Purcell enhancement have a mature theory of *physical upper bounds* — given a material and a volume, how large can the response be [22]. To our knowledge (as of 2026), **no bound theory of Chao type exists for F_Q over structures**; the closest results are the disorder-average invariance [14] and the FI continuity equation [15]. At the same time, F_θ hands inverse design a ready-made objective function — maximize the top eigenvalue of Eq. (3.2) over *structures*, not states. (One adjacent debate already proved the stakes: exceptional-point sensors show diverging *response*, yet for the linear, reciprocal EP sensors analysed so far QFI analysis found the noise diverges in step — classical response is not quantum estimability, and QFI is the final judge [25, 26] — though an EP accelerometer with enhanced signal-to-noise ratio [27] shows the last word is not in; extending the verdict to engineered nanophotonic environments is a sister problem to the seeds below.) Three concrete openings, each sized for a master's thesis:

1. **Inverse design for QFI — the quantum structure-end problem** (numerical + theory; pipeline: FMQA, factorization machine with quantum annealing). The tutorial solved "given the structure, optimize the state." The classical version of the dual — *given the estimation target, optimize the nanostructure* — has just been opened [23, 24]; its QFI version is open: use the top eigenvalue of F_θ (classical warm-up, then QFI) as the objective of topology optimization, and ask whether a Chao-type upper bound exists for it [22, 13].

2. **Quantum rulers on topological light textures** (theory + numerics). Displacement and rotation estimation in superoscillatory, singular, and skyrmionic fields: where does quantum light beat the classical FI of the picometric rulers [11, 12], in the lineage of multimode quantum metrology [19, 20]?

3. **Generator engineering with embedded emitters** (theory + numerics). The entire program above probes with *injected* light. Move the probe *inside*: for an emitter embedded in the structure, the generator G of Eq. (2.3) is dictated by the vector LDOS, transverse spin, and chiral coupling of the local modes [21] — the state-end-to-emitter-end shift of the whole field, and this lab's home advantage.

Each of these has only a sparse literature — that is the point. The groups that proved "the limit is a contract" still negotiate on behalf of the light; someone will negotiate for the structure, and for the emitters inside it. Join now, and you are not entering a field; you are founding one.

In-class discussion 3. *If the coupling — and therefore F_Q — can be designed, is the "quantum limit" still a limit?* Open question, on purpose. The SQL fell to squeezing; the Heisenberg limit is a statement about $(\Delta G)^2$ at fixed resources — but who fixes the resources when the generator itself is designable? — the doorway to the seeds above.

Summary: Section 3

F_Q traces back through the coupling to the vacuum mode structure — which nanophotonics designs, Eq. (3.1); Purcell was the prototype. Classically the case is closed: picometric rulers (Zheludev) and the eigenvalue problem $F_\theta = \partial_\theta S^\dagger \partial_\theta S$ (Rotter), with information flow steerable and enhancement provably a matter of design, not complexity. Everything so far optimizes the light; the classical structure end has just been opened, while the quantum structure end (QFI-targeted inverse design, unknown upper bounds) and the emitter end (generator engineering) are open — the three thesis seeds live exactly there.

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